Name_____ Date_____ Period_____

LESSON 9-3 · HOMEWORK PACKET · SUPPORTED · SESSION 1 · PAGES 1–2

Equations from Tables

Maryland Standard 6.EE.C.9 · Learning target – I can write an equation that matches a table of values.

REMEMBER

- An **equation** like $y = 5x$ is a rule that works for **every** row.
- Same $y \div x$ in every row: y = (that number) × x. Same $y - x$: $y = x$ + (that number).
- Check your equation with **two** rows.

WORKED EXAMPLE

Write an equation for x: 2, 3, 5 and y: 12, 18, 30.

Step 1 $12 \div 2 = 6$, $18 \div 3 = 6$, $30 \div 5 = 6$

Step 2 Check: $6 \times 5 = 30$

Answer $y = 6x$

WORD BANK

equation – a rule with an equal sign, like y = 5x

rate – the amount for each 1

check – test your rule on another row

PRACTICE

show your work · label units

1. Write an equation for each table.

Step 1 Divide each y by its x. Same number every time? Write y = (number)x.

Step 2 If not, find $y - x$. Same number? Write $y = x$ + (number).

a. x: 1, 2, 3 y: 7, 14, 21 $y =$ _____

b. x: 1, 2, 3 y: 10, 11, 12 $y =$ _____

c. x: 2, 4, 6 y: 10, 20, 30 $y =$ _____

d. x: 5, 6, 7 y: 2, 3, 4 $y =$ _____

e. x: 0, 4, 8 y: 6, 10, 14 $y =$ _____

f. x: 1, 3, 5 y: 9, 27, 45 $y =$ _____

g. x: 2, 3, 4 y: 16, 24, 32 $y =$ _____

h. x: 10, 12, 14 y: 5, 7, 9 $y =$ _____

2. Write the equation. Then complete the table.

Step 1 Divide 22 by 2 to find the rate.

Step 2 Use the rate to fill in the empty boxes.

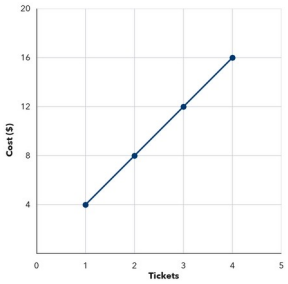
x	2	3	4	6	9
y	22	33	44		

Equation_____

3. Stretch. A table shows x: 2, 5, 9 and y: 0, 3, 7. Write an equation. Then find y when $x = 20$.

Answer_____

4. Tickets. The graph shows the cost of fair tickets. Write an equation for the graph. Then find the cost of 10 tickets.



Step 1 Read one point on the graph, like (1, 4).

Step 2 Divide the cost by the tickets. Then multiply by 10.

Answer_____

5. Find the Error. For x: 2, 4, 6 and y: 6, 8, 10, Owen wrote $y = 3x$ because $3 \times 2 = 6$. Explain his mistake and write the correct equation.

Frame *Owen's equation only works for ___. The correct equation is y = ___.*

_____

_____

CHECK YOURSELF	☐ I can do this on my own	☐ I need a little help	☐ I have a question about #____

AT HOME Make a 3-row table for a rule. Ask someone at home to find your equation.

Name_____ Date_____ Period_____

LESSON 9-3 · HOMEWORK PACKET · SUPPORTED · SESSION 2 · PAGES 3–4

Equations from Situations

Maryland Standard 6.EE.C.9 · Learning target – I can write an equation from a described situation and identify its variables.

REMEMBER

- A **rate** ("each," "per") multiplies x. A **starting amount** or fee is added once.
- y = (rate) × x + (starting amount). Check it with one value from the story.

WORKED EXAMPLE

A gym charges \$10 to enter plus \$7 per hour. Write an equation for the cost y of x hours.

Step 1 Rate: 7 per hour, so $7x$. Fee: add 10.

Step 2 Check 3 hours: $7 \times 3 + 10 = 31$

Answer $y = 7x + 10$

WORD BANK

rate – "each" or "per"; it multiplies x

fee – a cost paid one time; it is added

starting amount – what you have at the start; it is added

PRACTICE

show your work · label units

1. Write an equation for each situation.

a. Each pack holds 12 pencils. Pencils y in x packs. y = _____

b. Jamal is 3 years older than his sister. Jamal's age y when she is x. y = _____

c. A club costs \$20 to join plus \$15 a month. Cost y for x months. y = _____

2. **Bowling.** Bowling shoes cost **\$4** one time. Each game costs **\$5**. Write an equation for the cost y of x games. Find the cost of 4 games.

Step 1 Multiply the games by \$5.

Step 2 Add the \$4 shoe fee one time.

Answer_____

3. **Taxi.** A taxi charges \$3 to start plus \$2 per mile. Write an equation for the cost y of x miles. Find the cost of 7 miles.

Step 1 Multiply the miles by \$2.

Step 2 Add the \$3 start fee one time.

Answer_____

4. Car Wash. A class has $30 and earns $8 per car washed. Complete the table. Write an equation.

Step 1 At $x = 0$, no cars are washed yet. What does the class have?

Step 2 Add $8 for each car.

Cars (x)	0	1	2	5
Total $ (y)				

Equation_____

5. Find the Error. For "a $25 fee plus $10 an hour," Nia wrote $y = 35x$. Explain her mistake and fix it.

Frame *Nia's mistake is ___. The fee is paid ___, so y = ___.*

_____

_____

TEST PRACTICE

choose the best answer

6. A plumber charges a $40 fee plus $25 per hour. Which equation gives the cost c for h hours?

A $c = 40h + 25$ **B** $c = 25h + 40$ **C** $c = 65h$ **D** $c = 25h$

7. Which situation matches $y = 3x + 12$?

A 3 cans per box, 12 boxes

B Start with $3, earn $12 per hour

C Start with 12 gal, add 3 gal per min

D $12 for 3 people

CHECK YOURSELF	☐ I can do this on my own	☐ I need a little help	☐ I have a question about #___